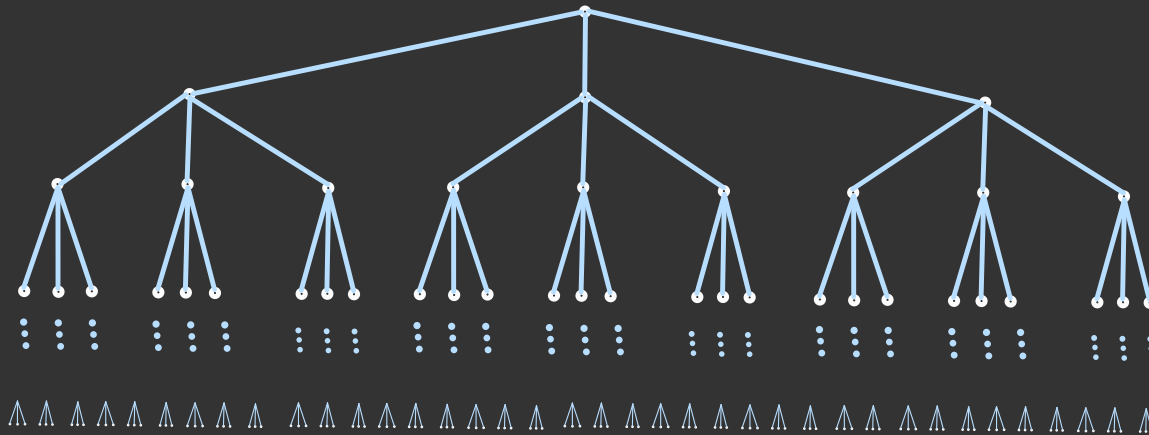


runtime of
bfs and dfs

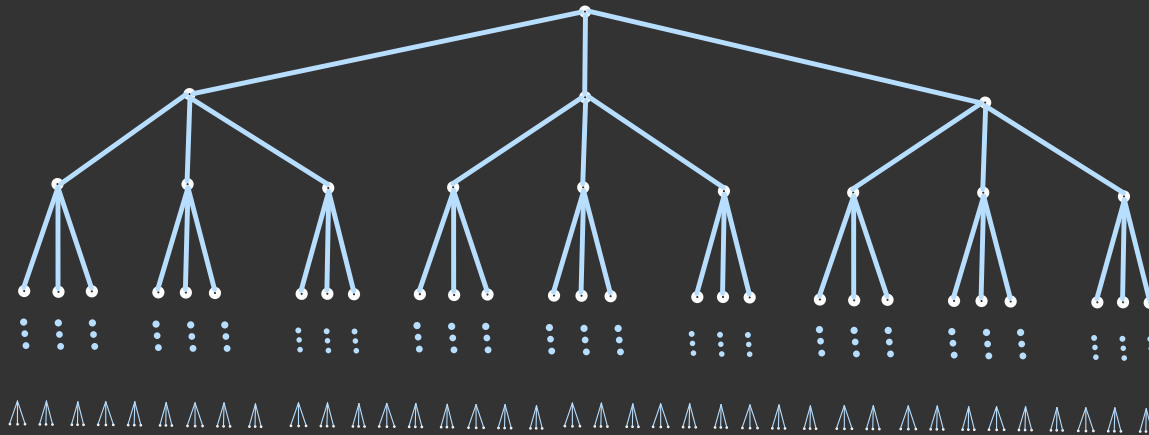
CSCI
270

in a full tree with branching factor b ,
how many nodes are at depth d ?



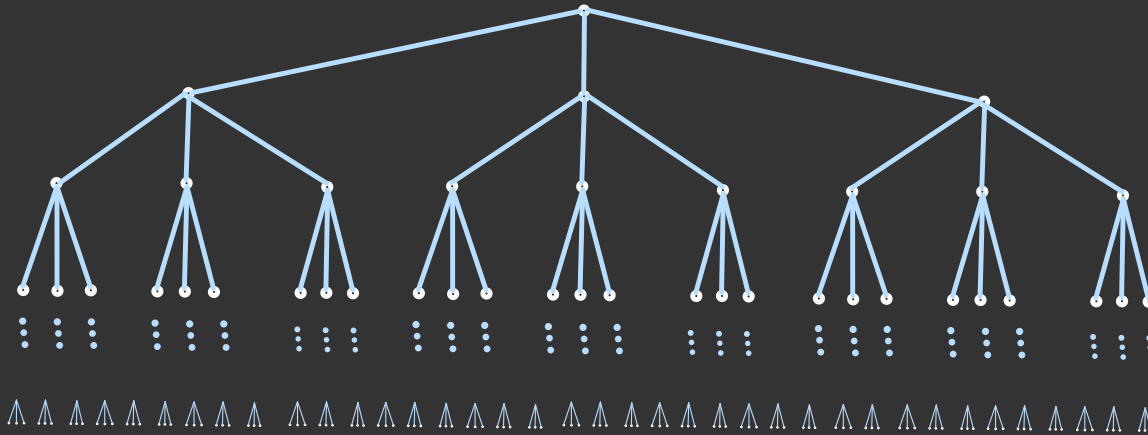
depth	# nodes
0	?
1	?
2	?
3	?
$\vdots$	$\vdots$
d	?

in a full tree with branching factor b ,
how many nodes are at depth d ?



depth	# nodes
0	1
1	b
2	b^2
3	b^3
$\vdots$	$\vdots$
d	b^d

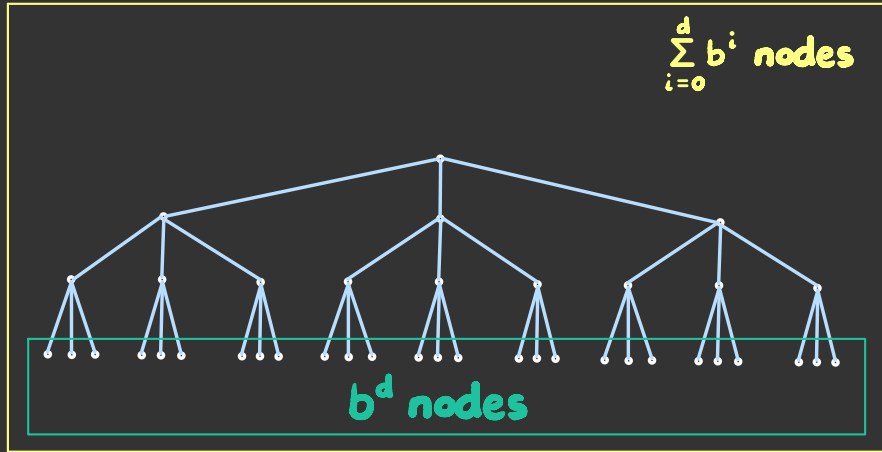
so a full tree with branching factor b
and depth d has $\sum_{i=0}^d b^i$ nodes in total



depth	# nodes
0	1
1	b
2	b^2
3	b^3
$\vdots$	$\vdots$
d	b^d

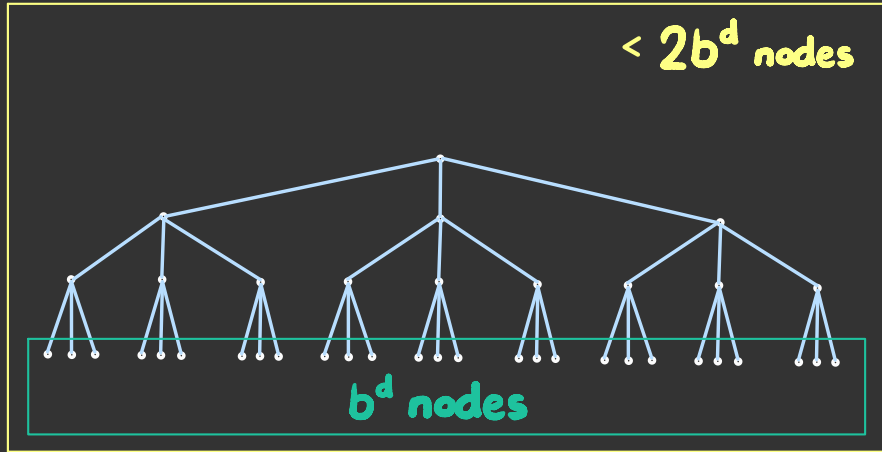
fun fact: if $b \geq 2$, then $\sum_{i=0}^d b^i < 2b^d$

what does this
imply about the
number of
internal nodes
(non-leaves)?



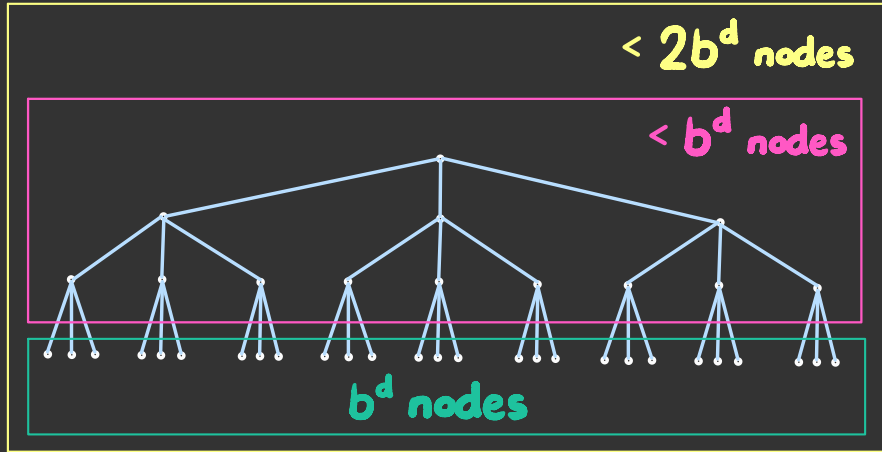
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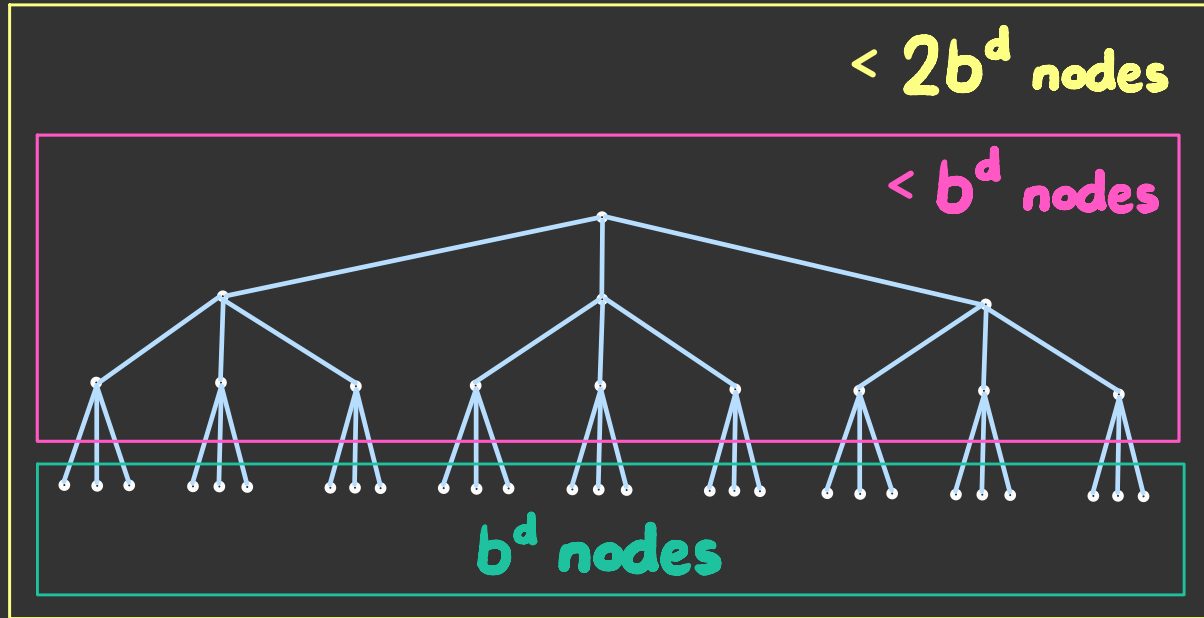


fun fact: if $b \geq 2$, then $\sum_{i=0}^d b^i < 2b^d$

what does this
imply about the
number of
internal nodes
(non-leaves)?



trees with branching factor $b \geq 2$
have more leaves than internal nodes



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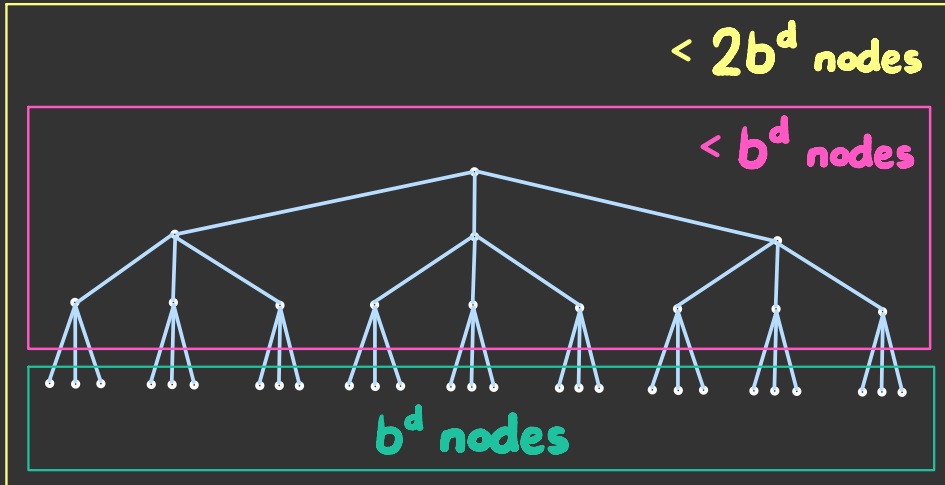
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THE SCIENCES

Fact or Fiction?: Living People Outnumber the Dead

Booming population growth among the living, according to one rumor, outpaces the dead

By Clara Curtin on March 1, 2007



Everybody feels they are right- some are confident, some are open. I'm definitely open to all of these possibilities, but science has it's say(irriducible complexity, creation- i.e. Einsteins clearly stated belief for existence, micro vs macro evolution). Personally that say has led me to be more inclined to the 6000 years theory (and this is ALL THEORY) but I remain open. Thus, if I had to choose today I'd say there are more alive today than ever before. But I'm (hopefully) always open for more info to bring more fact/truth

Jamie, Thorold Canada

No, there are definitely more people alive than dead. That is FACT!

Pippy Van't-Scmit, Kloofenhoffen Namibia

There are more dead people today than living.

Abdulai Ibrahim, Tamale Ghana

6000 years? Really? I thought it was just some backward bible belt Americans who did not believe in evolution. Flat earth next maybe eh?

Barry Ley, Dubai UAE

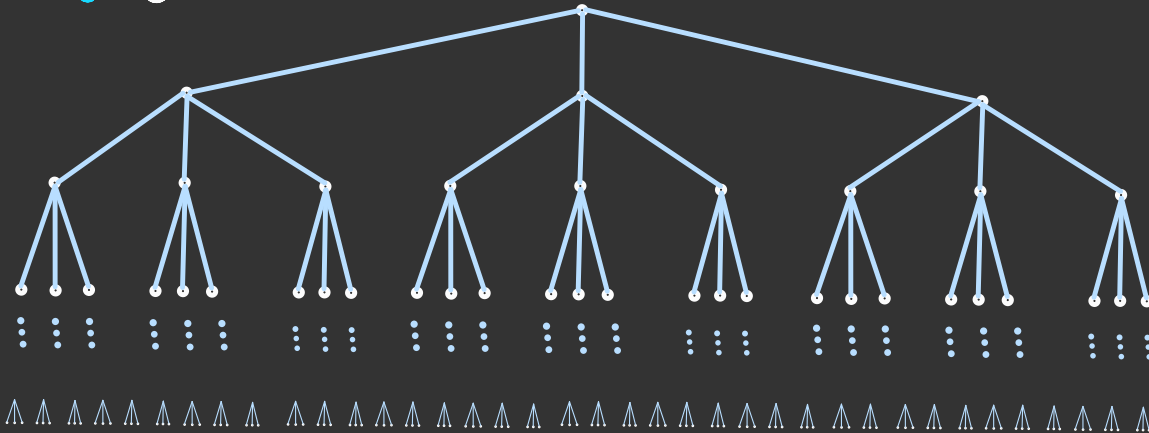
Did some actually state the earth at 6000 years old? Why aren't they dead from stupidity? Ugh. Whatever the number is, there are a lot more brain dead people than alive.

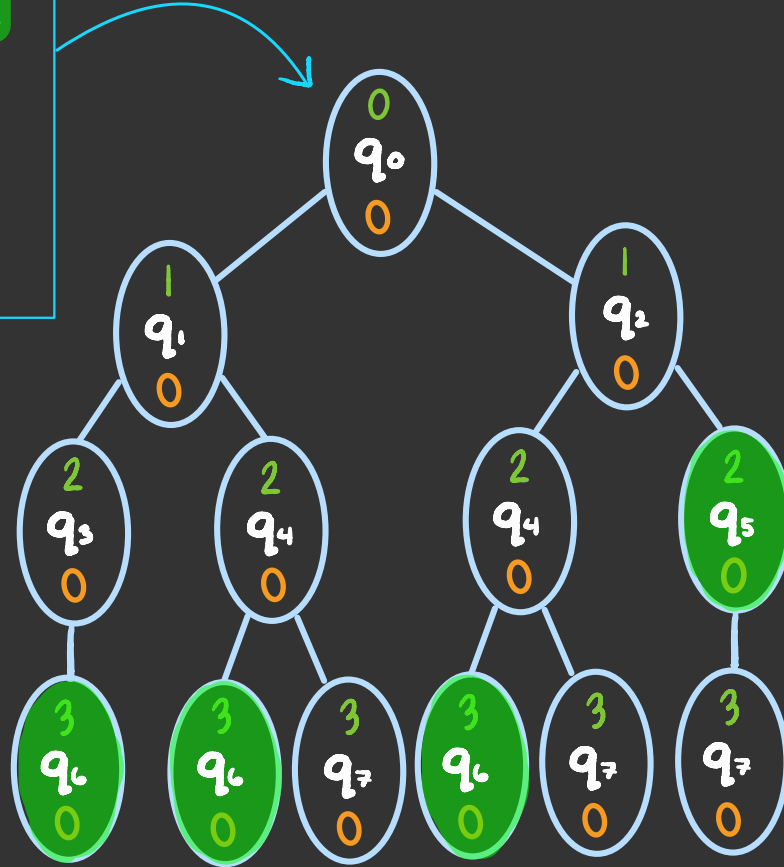
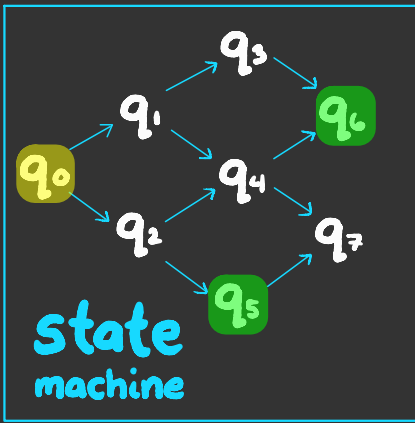
Tom, New York United States

so a full tree with branching factor $b \geq 2$

and depth d has a total of

$$\sum_{i=0}^d b^i < 2b^d = O(b^d) \text{ nodes}$$





solution depth

d

the length of the shortest search path that ends at a final state

maximum depth

m

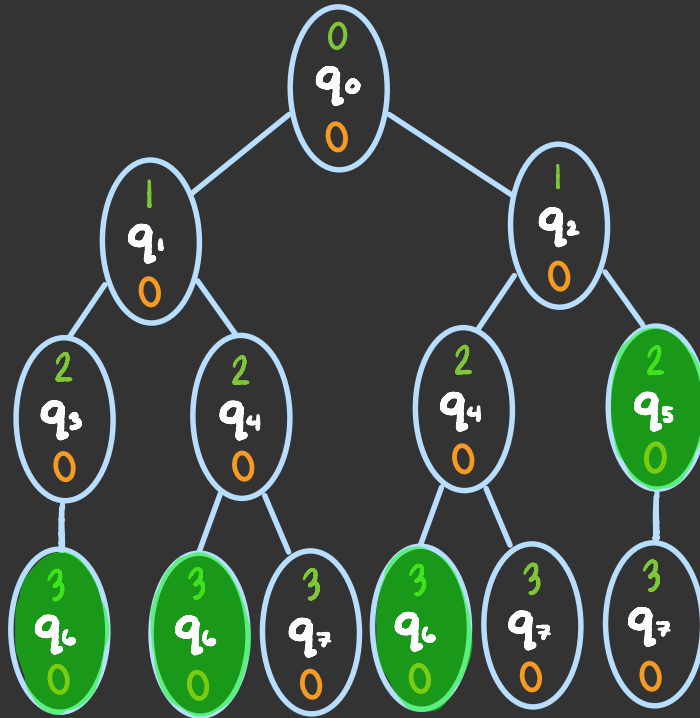
the length of the longest search path

branching factor

b

the maximum number of successors of a state

what is the worst-case runtime of bfs?



solution depth

d

the length of the shortest search path that ends at a final state

maximum depth

m

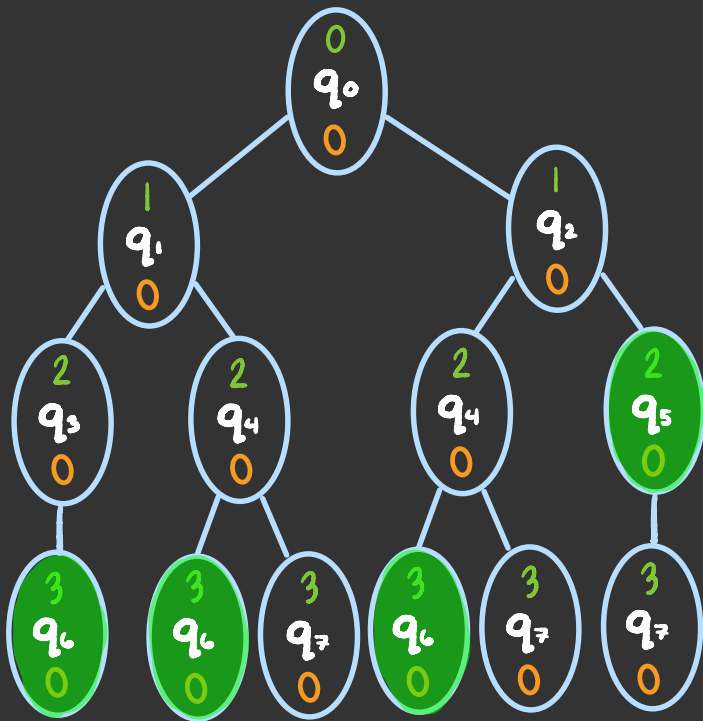
the length of the longest search path

branching factor

b

the maximum number of successors of a state

recall: for $j < k$, every node at search depth j is visited before any node at search depth k



solution depth

d

the length of the shortest search path that ends at a final state

maximum depth

m

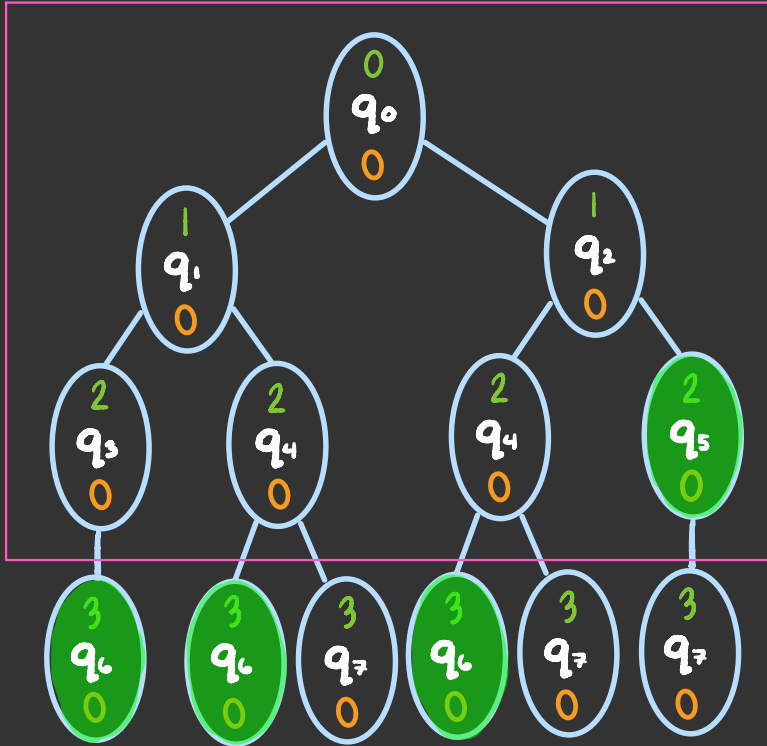
the length of the longest search path

branching factor

b

the maximum number of successors of a state

thus it visits $O(b^d)$ nodes



solution depth

d

the length of the shortest search path that ends at a final state

maximum depth

m

the length of the longest search path

branching factor

b

the maximum number of successors of a state

what is the worst-case
runtime of dfs?

solution depth

d

the length of the shortest search
path that ends at a final state

maximum depth

m

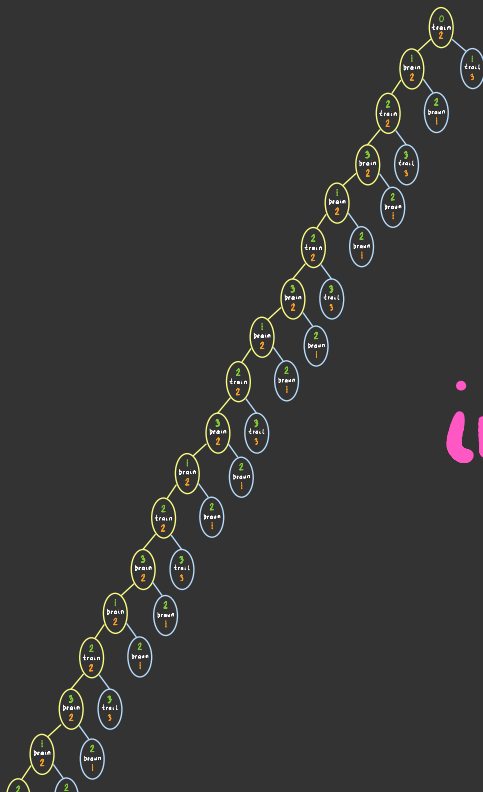
the length of the
longest search path

branching factor

b

the maximum number of
successors of a state

what is the worst-case runtime of dfs?



infinite!



a.k.a.
 $O(\text{😭})$

solution depth

d

the length of the shortest search path that ends at a final state

maximum depth

m

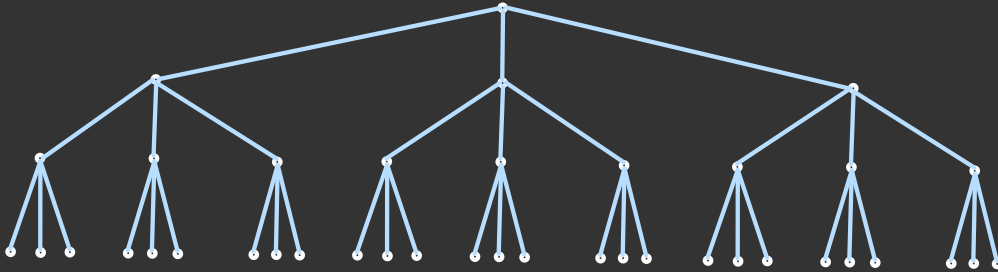
the length of the longest search path

branching factor

b

the maximum number of successors of a state

but what if the search tree
has **finite depth**?



there might not be a solution,
or it might be in the last place we look

solution depth

d

the length of the shortest search
path that ends at a final state

maximum depth

m

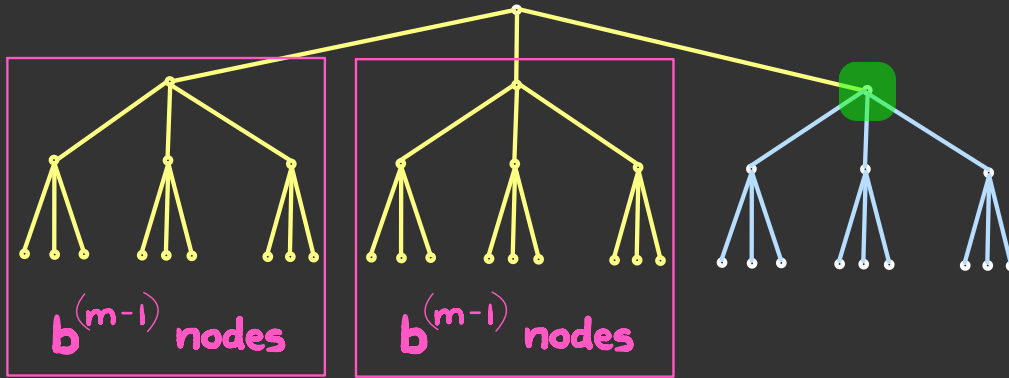
the length of the
longest search path

branching factor

b

the maximum number of
successors of a state

even if solution depth $d=1$,
we might visit $(b-1)b^{(m-1)}$ nodes



thus dfs can visit $O(b^m)$ nodes

solution depth

d

the length of the shortest search path that ends at a final state

maximum depth

m

the length of the longest search path

branching factor

b

the maximum number of successors of a state